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Dr. Manisha Aswal
Scientific Handling Editor
Array
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Dear Dr. Aswal,

Please find enclosed the revised version of our manuscript ARRAY-D-26-01253, **"Attention-Aware Task Scheduling via Deep Reinforcement Learning under Partial Observability for Human-Centric Productivity Optimization"**. We thank you and the two reviewers for constructive comments, which identified real weaknesses of the original submission. We have used them to revise the study itself, not only the text: all experiments were redesigned and re-run, new experiments were added, and the manuscript was rewritten. The author list is unchanged.

The main revisions are:

- **Novelty and positioning** (Editor, Reviewer 4). A new paragraph in the Introduction states the four elements that distinguish the work from fatigue-aware scheduling and human–robot task allocation, and the related work and positioning table were extended with recent human-aware scheduling studies.
- **Stronger baselines and a second DRL algorithm** (Reviewer 4). The study now compares a value-based learner (AA-DQN) and a new policy-gradient learner (AA-PPO) with nine rule-based schedulers, five of which are grid-tuned and cognition-aware, and with a vanilla DQN.
- **Rigorous evaluation** (Reviewer 4). Ten training seeds per main configuration, 500 held-out test days with common random numbers, disjoint seed sets for training, validation, tuning and testing, bootstrap confidence intervals, and Holm-corrected Wilcoxon tests with effect sizes. In total, 114 agents were trained.
- **Reward-independent metrics** (Reviewer 4). Conclusions are based on the on-time rate, deadline violations, completion rate and a difficulty-weighted (workload-adjusted) on-time rate, with a control agent trained without cognitive reward terms.
- **Complete specification** (Reviewers 3 and 4). Transition rules, dispatch function and deadline logic are formalised, all parameters are listed with their sources, and all hyperparameters and the way every parameter was set are reported.
- **Sensitivity analyses** (Reviewer 3). New experiments vary the learning rate, discount factor, history length, network components and reward weight, and each of the seven cognitive parameters by $\pm 30\%$.
- **Corrected conclusions** (Reviewer 4). The original claim that learning helps only when cognition is modelled was contradicted by the new controlled ablations and has been retracted.
- **Presentation** (Editor, Reviewer 3). A Nomenclature, numbered equations, 13 recent references (including the one suggested), a discussion of advantages and disadvantages, an extended limitations section, a graphical abstract, five highlights, and the Ethics, Data availability, CRediT, Competing interest, Funding and AI-use statements. The manuscript was proofread throughout.

With the stronger protocol, the results are more modest than in the original draft but much more trustworthy. Rules that never rest perform poorly (EDF completes 46.8% of tasks on time), and tuned rest rules are strong. The best learner, AA-PPO, significantly exceeds the best of nine tuned rules in return (13.04 versus

12.30) and on-time completion (75.1% versus 71.4%). We also report where learning does not help, notably on unseen workloads.

We enclose a point-by-point response to all comments, a highlighted version of the manuscript, in which the passages that answer a comment are printed in blue and tagged with the comment number, a clean version, the revised supplementary material, the graphical abstract and the highlights. Because nearly every section was rewritten, a word-level comparison would mark almost every sentence; the tagged highlighting is intended to make the responses easy to locate.

We confirm that the manuscript is original, has not been published previously and is not under consideration elsewhere, and that all authors have approved the revised version.

Yours sincerely,

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